SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10 **Total Marks:20**

Annexure A

CLASS TEST

Criteria	Understanding of Task *(2)*	Procedure & Execution *(2.5)*	Observation & Result *(2)*	Viva Voce / Conceptual Response *(2)*	Time Management & Presentation *(1.5)*	*Total(10)*
Weightag e	*20%*	25%	20%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						
Q10						

Code of Conduct:

I S. Chandra Shekhar / 192372163 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *S. Chandra Shekhar*

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS: **Total Marks: 10x2 =20**

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formals[L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set x = {4,5,6,7,9}, y = {8,3,7,5,6}

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, l5]

CLASS TEST - ANSWER SHEET (as submitted)

CO2_ATL-3

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Test-1

① Define Data mining and list any two basic data mining task. Explain the steps involved in KDD process and justify the roles of data preprocessing

Sol

Data Mining

Data Mining is the process of extracting useful pattern, trends and knowledge from large datasets.

Two basic data Mining tasks:-

1. Classification
2. clustering

KDD (knowledge Discovery in Databases)

The Data Selection

2) Data cleaning
3) Data integration
4) Data transformation
5) Data mining

Role

It removes missing values, noise and inconsistencies, improving the accuracy and reliability of mining result

② What is the data cleaning? Mention two common data quality issue. Explain method to handle missing value and Noisy data.

Data Cleaning is the process of detecting and correcting inaccurate incomplete or inconsistent data.

Two data quality issues

1. Missing values
2. Noisy data

Handling Missing Values

- Replace With mean/Median/mode
- Predict Missing values using machine learning

Handling Noisy data

- Binning
- Regression
- clustering.

Example

Missing age can be replaced with average age.

(3) Data integration and list two challenges assicoated with it Explain how redunancy and inconsistency are handled.

Sol Data integration

Combined data From Mutiple Source into a single unifed dataset

Challenges

1. Data redunancy
2. Data inconsistency

Handling redundancy

- Remove duplicate records
- Use Correlation analysis

Handling inconsistency

- Standarlize formate
- Apply data cleaning

(4) What is Covariance Analysis? List the two covariance formula?

Sol

Convariance Analysis

it Measures how to two variables change together

Population covariance

$$Cov(x,y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Sample Covariance

$$Cov(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

(5) Suppose two stock A and B have the following value in one week (2,5) ; (3,8) (5,10) (4,10) , (6,4) question :- If stock are affected by same industry trends, will their prices rise or fall together and independent?

Sol

A = (2,3,5,4,6)

B = (5,8,10,11,14)

Since both stock prices generally increase together the covariance is positive

Conclusion

The Stocks tend to rise together and are Not independent

(6) Explain the architecture of type of data mining system with Neat diagram Discuss the role of each Component and analyze how they interact to support knowledge discovery

sol

Components

1. Data Sources
2. Database / Data warehouse
3. Data cleaning
4. Data mining Engine
5. Pattern Evaluation module
6. User interface

Working

Data is collected → cleaned → Store → mined → evaluated → usefull is presented to user

(7) Find covariance for following data set $X = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$

sol

$x = (4, 5, 6, 7, 9)$

$y = (8, 3, 7, 5, 6)$

mean $x = 6.2$

mean $y = 5.8$

$$\text{cov}(x,y) = \frac{\sum (x-\bar{x}) \cdot (y-\bar{y})}{n-1}$$

Covariance $= -0.075$

(9) A hospital is analyzing patient age data to identify broad health trend. Because the raw data contians minors variations and noise the the analyst need to perform local smooth equal width and equal frequency (13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70)

sol

Major preprocessing Steps

1. Data Cleaning
2. Data integration
3. Data transformation
4. Data Reduction
5. Data Discretization

Pipeline

Raw Data → Cleaning → integration → Transformation → Reduction → mining

The sequence improve data quality mining.

(8) Describe the major step in pre-processing. Given a dataset with missing values, outerliers and inconsistent formats propose they interact to Support Sy pipeline and justify the sequence of steps

8) Equal frequency bining divides data into bin with nearly equal Number of values

Bin 1

13, 15, 16, 16, 19, 20, 20, 21, 22

median = 19

Smooth values

19, 19, 19, 19, 19, 19

Similary smooth the remaining bin using their median

Equal - frequency bining reduce Noise.

(10) Explian data discretization and its importance in data mining compare equal width and equal frequency bining with suitable Example and evaluate their impact on data analysis.

sol

Data Discretization

Converts continous values into discrete intervals (bins)

equal width	equal frequency
1) equal interval size	1) equal No of records
2) Simple to implement	2) Handles skewed data better
3) may Create uneven bins	3) Balanced distribution

Example

Data = 10, 20, 30, 40, 50, 60

equal width

10 - 26

27 - 43

44 - 60

equal - Frequency

- (10, 20)

(30, 40)

(50, 60)